Deep Feature Correlation Learning for Multi-Modal Remote Sensing Image Registration

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***Abstract*— Deep descriptors have advantages over handcrafted descriptors on local image patch matching. However, due to the complex imaging mechanism of remote sensing images and the significant differences in appearance between multi-modal images, existing deep learning descriptors are unsuitable for multi-modal remote sensing image registration directly. To solve this problem, this article proposes a deep feature correlation learning network (Cnet) for multi-modal remote sensing image registration. First, Cnet builds a feature learning network based on the deep convolutional network with the attention learning module, to enhance feature representation by focusing on meaningful features. Second, this article designs a novel feature correlation loss function for Cnet optimization. It focuses on the relative feature correlation between matching and nonmatching samples, which can improve the stability of network training and decrease the risk of overfitting. In addition, the proposed feature correlation loss with a scale factor can further enhance network training and accelerate network convergence. Extensive experimental results on image patch matching (Brown, HPatches), cross-spectral image registration (VIS-NIR), multi-modal remote sensing image registration, and single-modal remote sensing image registration have demonstrated the effectiveness and robustness of the proposed method.**

***Index Terms*— Attention mechanism, feature correlation, image registration, multi-modal image, remote sensing image.**

I. Introduction

REMOTE sensing image registration is the process of geometrically aligning a reference image and a target image, which show the same scene and maybe obtained by different sensors, at different times, from different viewpoints, or under different illumination conditions. Remote sensing image registration is widely used in various applications, such as change detection [1], multi-modal image fusion [2], and object detection [3]. In particular, multi-modal image registration is essential to fuse rich and complementary information between multi-modal remote sensing images. However, multi-modal remote sensing image registration (optical–SAR) is more challenging than single-modal remote sensing image registration (optical–optical, SAR–SAR). There are significant differences caused by different imaging mechanisms in multi-modal remote sensing images, which will increase the challenges of image registration greatly [4], [5]. Moreover, the speck noise in SAR images [6] will also affect the performance of image registration.

The remote sensing image registration methods can be roughly divided into intensity-based methods, transform-domain-based methods, and feature-based methods. The intensity-based methods aim to determine the optimal transformation parameters by a specific similarity metric based on pixel information, such as mutual information (MI) [7], normalized cross correlation (NCC) [8], and the sum of square differences [9]. However, such methods are limited to images with small differences in intensity. The transform-domain-based methods transfer the image aligning task from the spatial domain to the frequency domain by Fourier transform [10]. They are more suitable for matching images with translation transformation but ineffective for great viewpoint changes. The feature-based methods are more robust for remote sensing image registration. They aim to build correspondences across images based on local features. The traditional feature-based methods rely on handcrafted local descriptors, such as the difference in Gaussian (DOG) and statistical information of the image gradient. The representative local descriptors are scale-invariant feature transform (SIFT) [11] and its variants, such as speeded-up robust feature (SURF) [12], Affine-SIFT [13], and OS-SIFT [14]. These gradient-based methods are invariant to scale and rotation transformations. However, their robustness and generalization are insufficient to meet the needs for accurate multi-modal remote sensing image registration. To extract the common feature between multi-modal images, Ye *et al.* [15] proposed to extract the shape properties based on the dense local self-similarity or structural properties based on the histogram of orientated phase congruency [5], and then search the optimal transformation parameters based on the template matching strategy. However, the discrimination of shape features needs to be further improved in local feature-based image registration.

Recently, with the success of deep learning in various image processing tasks, deep descriptors have shown significant advantages over handcrafted descriptors in image matching. The advantages of deep descriptors on image matching and

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registration mainly come from the multiple nonlinear mapping network structure and the learnable mechanism. First, the deep convolutional network can extract high-level features through multiple nonlinear mapping layers, which are more robust to speckle noise and image appearance changes than handcrafted descriptors. Second, the deep convolutional network is learnable and trainable according to the optimization loss, which expects the matching features to be similar while the nonmatching features are dissimilar. The learnable mechanism encourages the deep convolutional network to automatically extract invariant features to various changes, such as illumination changes, rotation transformation, and even image modality changes. Meanwhile, it can extract the discriminative features to separate matching image patches from many nonmatching image patches, improving image matching and registration performances. Therefore, the deep descriptor learning-based image registration performances mainly depend on the network structure and network optimization. Specifically, previous works [16]–[20] proposed to use the deep convolutional network to learn high-level features from local image patches. Most of them built local correspondences between images based on the feature distance (Euclidean distance). The deep descriptor learning network can be optimized by the feature-distance-based loss, such as contrastive loss [17] and triplet loss [20]. The contrastive loss aims to minimize the feature distance of matching patch pairs and maximize the feature distance of nonmatching patch pairs. Unlike contrastive loss paying more attention to the absolute feature distance of patch pairs, triplet loss mainly focuses on the relative feature distance. It constrains the feature distance of nonmatching patch pairs to be larger than that of matching patch pairs by a margin. These deep descriptor learning methods were initially proposed for natural image matching. Due to the complex imaging mechanism of remote sensing images, the existing deep descriptor learning methods are unsuitable for multi-modal remote sensing image registration directly. For example, contrastive loss is too strict for matching image patches with significant differences and transformation. It expects the feature distance of matching patches to be as close as possible to zero, which will increase the risk of model collapse [21]. Moreover, it is not easy to find the optimal distance margin in triplet loss. Ideally, the optimal margin should increase dynamically with the learning of the matching network rather than being fixed.

To solve the above problems, this article proposes a deep feature correlation learning network (Cnet) for multi-modal remote sensing image registration, which improves the deep descriptor learning network from network structure and network optimization. First, this article builds a deep convolutional network with an attention module (AM) to extract high-level features from images. On one hand, compared with the fixed handcrafted feature extraction operator, the deep feature learning network is more flexible, automatically learning the high-level feature through network optimization. On the other hand, AM will encourage the feature learning network to focus on meaningful high-level features, which are invariant to illumination changes, rotation transformations, image modality changes, and more discriminative to identify nonmatching patches. Second, this article proposes a novel feature correlation loss function to optimize the feature learning network, which can improve network performance, enhance network training stability, and accelerate network convergence. Specifically, it constrains the feature correlation of matching patch pairs to be larger than that of nonmatching patch pairs, which can relax the requirement for matching patches and tolerate the differences between matching image patches. Meanwhile, the proposed optimization loss function includes a scale factor to enhance network training and accelerate network convergence.

The main contributions of this article can be summarized as follows.

1) This article proposes a deep feature Cnet for multi-modal remote sensing image registration. With the help of the deep convolution network and attention learning module, Cnet can exploit invariant and discriminative features to improve the performances of deep descriptors on image patch matching and image registration.
2) This article designs a novel feature correlation loss function to optimize the network, which can improve the matching performance, enhance the network stability in the training process, and accelerate network convergence.
3) Extensive experimental results on image patch matching, cross-spectral image registration, multi-modal remote sensing image registration, and single-modal remote sensing image registration show the effectiveness and robustness of our method.

The rest of the article is organized as follows. Section II presents related work on remote sensing image registration and attention mechanism. Section III introduces our method and gives the technical details. The experiments are described in Section IV. A summary is presented in Section V.

II. Related Work

This section introduces related works on remote sensing image registration methods, deep-learning-based image matching methods, and attention mechanisms.

A. Remote Sensing Image Registration

Remote sensing image registration is very important for dealing with inevitable geometric and radiometric distortions during the process of acquiring the data [22]. The traditional remote sensing image registration methods can be divided into three classes: intensity-based methods, transform-domain-based methods, and feature-based methods.

The intensity-based methods determine the optimal transformation parameters by maximizing the image similarity based on image pixel information. Maes *et al.* [7] proposed a multi-modal image registration method by maximizing MI. Sarvaiya *et al.* [23] proposed to align images by template matching based on NCC. Liang *et al.* [8] proposed to search the matching local regions with the maximal spatial relations and MI (SMI) by ant colony optimization. However, intensity-based methods are sensitive to the changes in pixel intensity. Their image registration performance will be affected by

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illumination change and image noise. The transform-domain-based methods compute transformation parameters in the frequency domain. Reddy and Chatterji [10] proposed a fast-Fourier-transform-based method to find the optimal transformation parameters.

Compared with raw image pixel intensity, local features are more stable and robust to image changes. So, the widely used remote sensing image registration methods are based on local features. Initially, local image features are related to the statistical information of image gradients, including gradient amplitude and gradient direction. The most popular local descriptor is SIFT [11]. It obtains 128-dimensional descriptors based on the histogram of local gradient information. To improve the speed of SIFT, SURF computes 64-dimensional descriptors based on the histogram of the Haar wavelet response [12]. Rublee *et al.* [24] proposed a fast binary descriptor, ORB. Yu and Morel [13] proposed an affine invariant SIFT (Affine-SIFT). It simulates the parameters of scale, camera longitude, and latitude angles and normalizes image translation and rotation. In addition, to decrease the negative influence of speckle noise in synthetic aperture radar (SAR) images, several articles have modified the original SIFT. For example, Schwind *et al.* [25] proposed SIFT-octave (SIFT-OCT) to modify the keypoint detection step, which reduces the influence of noise by skipping the first octave of the scale space. Wang [26] built the anisotropic scale space using a bilateral Gaussian filter. Dellinger *et al.* [27] proposed SAR-SIFT to detect the keypoint based on a multiscale Harris operator and design new gradient features that are robust to speckle noise. Shen and Bao [28] combined NCC and SIFT to perform fast and accurate remote sensing image registration. Goncalves *et al.* [29] combined image segmentation and SIFT to achieve accurate points for remote sensing images. To enhance the robustness of image registration, Gonçalves *et al.* [30] proposed a histogram-based image segmentation (HAIRIS) method for automatic image registration. Bhatt and Padmanabhan [22] introduced the development of operational software for multitemporal and multiband image registration. In addition, to improve remote sensing image registration accuracy, Ye *et al.* [31] proposed a novel keypoint detector, Harris–Laplace, which can simultaneously detect corners and blobs.

In the feature matching step, the commonly used matching strategy is the nearest neighbor (NN) matching strategy, which searches the matching point with the minimum distance; refer to the following equation:

$$j = \arg\min_{\forall k} \; D\left(f_i, f_k'\right) \tag{1}$$

where $D(f_i, f_k')$ is the feature distance between f_i and f_k', f_i is the feature of the ith keypoint in the first image, f_k' is the feature of the kth keypoint in the second image, and f_j' is the NN of f_i.

Due to local similarity of images caused by the repeatable local image pattern and the complex characteristics of remote sensing images, various false matching point pairs may disturb the performance of multi-modal remote sensing image registration. Thus, it is necessary to eliminate these false matching pairs. To eliminate potential false matching point pairs, previous researchers proposed the mutual NN (MNN) matching strategy, the NN threshold (NNT) matching strategy, and the NN distance ratio (NNDR) matching strategy [11], [32], which aim to find the matching pairs with high confidence. As shown in (2), MNN constrains the matching points are each others' NNs. For the ith keypoint in the first image, the feature distance between f_i and f_j' is minimum; meanwhile, for the jth keypoint in the second image, the feature distance between f_i and f_j' is minimum

$$\begin{aligned} j &= \arg\min_{\forall k} \; D\left(f_i, f_k'\right) \\ i &= \arg\min_{\forall k} \; D\left(f_k, f_j'\right) \end{aligned} \tag{2}$$

where f_j' is the NN of f_i, while f_i is the NN of f_j'.

NNT expects the feature distance of NN to be smaller than the distance threshold T; refer to the following equation:

$$\begin{aligned} j = \arg\min_{\forall k} \; & D\left(f_i, f_k'\right) \\ & D\left(f_i, f_j'\right) < T \end{aligned} \tag{3}$$

where T is the distance threshold.

NNDR requires the rate of the distance of NN to that of the second NN to be smaller than the ratio threshold RT; refer to the following equation:

$$\begin{aligned} j &= \arg\min_{\forall k} \; D\left(f_i, f_k'\right) \\ l &= \arg\min_{\forall k, k \neq j} \; D\left(f_i, f_k'\right) \\ & \frac{D\left(f_i, f_j'\right)}{D\left(f_i, f_l'\right)} < \mathrm{RT} \end{aligned} \tag{4}$$

where f_j' is the NN of f_i, f_l' is the second NN of f_i, and RT represents the ratio threshold.

In addition, Bian *et al.* [33] proposed a grid-based motion statistics method (GMS) to conduct smooth constraint based on the statistical number of matches in the support region. They believe that the true match neighborhoods will have more support matches than false match neighborhoods. So, GMS eliminates false matches where the number of matches in the support region is small. Fischler and Bolles [34] proposed the random sample consensus (RANSAC) method to find the optimal transformation matrix with the maximum number of support matching point pairs.

B. Deep-Learning-Based Image Matching

In recent years, deep learning neural networks have been introduced to image matching. MatchNet [16] proposed to use two convolutional neural networks with the same parameters (Siamese network) to extract features for image matching. Wang *et al.* [35] proposed a deep learning framework for remote sensing image registration. It uses a deep neural network to extract features and constructs training data through a self-learning method. Instead of treating all the samples equally, L2-Net [19] and HardNet [20] use a hard sample mining strategy to exploit effective samples for feature network training. Specifically, hard samples mean similar negative image patch pairs and dissimilar positive image patch

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pairs. These hard samples will provide strong penalize and effective gradient guidance for network optimization, which can increase the performance of descriptors and accelerate the network convergence. SOSNet [36] uses a second-order similarity regularization term to improve the generalization ability of descriptors. Softmargin [37] weights the loss value using the cumulative histogram of the difference between positive pairs and negative pairs, which is essentially a dynamic soft-margin process. To improve the convergence speed of network, Wang *et al.* [38] proposed an exponential loss for image patch matching. Quan *et al.* [39] proposed an aggregated feature difference learning network (AFD-Net) to fuse the feature difference from multiple levels and improve the local image matching results. Deshpande and Bhatt [40] proposed a Bayesian deep learning method for medical image registration, which estimates the displacement vector field of the moving image and undergoes grid resampling to warp the moving image using estimated DVF. In addition, Li and Fan [41] and Vos *et al.* [42] proposed a self-supervised image registration method that used the deep convolutional network to regress the spatial transformations by maximizing the similarity between the input image pairs. However, these deep-learning-based methods mainly focus on natural image matching and medical image registration. Due to the complex imaging mechanism, remote sensing image registration is more difficult. For example, the pixel value of remote sensing images is related to the sensor, object attributes, and surroundings. Moreover, significant differences between multi-modal images increase the difficulty of image registration. Thus, deep learning methods need to be further improved for multi-modal remote sensing image registration. This article proposes an effective deep learning network for more accurate multi-modal remote sensing image registration, which modifies the deep convolutional network with an attention modal and a novel feature correlation loss.

C. Attention Mechanism

The attention mechanism aims to explore meaningful features automatically to enhance their representation power and improve the final performance. It assigns a soft weight to each channel or spatial position of the feature maps. Specifically, important features have strong assignments while irrelevant features are suppressed [43], [44]. Attention mechanisms have enabled deep feature learning networks to achieve significant improvements in various computer vision tasks, such as image caption [45]–[47], semantic segmentation [48], [49], image classification [50], human pose estimation [51], image super-resolution [52], action classification [53], and person reidentification [54], [55].

Hu *et al.* [43] proposed a "squeeze-and-excitation" model to learn the interdependencies between the channels of feature maps. Woo *et al.* [44] proposed a "convolutional block AM" (CBAM) to perform attention learning along both the channel dimension and the spatial dimension of convolutional features, sequentially. Zhang *et al.* [56] introduced the self-attention model to the generative adversarial network and improved the performance of image generation. Wang *et al.* [50] combined attention mechanism and residual network for image classification. Zhang *et al.* [52] proposed a channel attention network for image super-resolution.

In this article, we present a feature extraction network based on the convolutional block AM. It can pay more attention to meaningful features to improve the performance of feature matching. In addition, we use feature correlation to build local correspondences between images. Moreover, we propose a novel feature correlation matching loss function to optimize the feature learning network, which can extract invariant features for matching samples and more discriminative features for nonmatching samples.

III. Method

A. Image Registration Pipeline

Image registration is the process of aligning two or more images. Formally, given two images $I_1(x, y)$ and $I_2(x, y)$, the image registration can be formulated as

$$I_2(x, y) = g(I_1(\phi(x, y))) \tag{5}$$

where ϕ represents the corresponding spatial transformation relationship between the two images, and g is the gray-scale interpolation operator.

In this article, we are mainly concerned with the spatial transformation relationship between image pairs, and the formulation is simplified to

$$I_2(x, y) = I_1(\phi(x, y)). \tag{6}$$

The feature-based image registration framework consists of five main steps.

1) *Keypoint Detection:* Keypoint detection aims to identify local interest points that are widely distributed on images and can be repeatedly detected even after different image changes, such as corners, Harris points, SIFT points, and ORB points. In this article, we use ORB to detect keypoints.
2) *Feature Extraction:* Generally, we use a feature vector to represent the local image patch. Ideally, the extracted feature should exhibit high invariance to transformations and high discrimination for nonmatching patches' identification.
3) *Feature Matching:* Feature matching builds local correspondences between image patch pairs based on the similarity of the extracted features, such as the nearest feature distance.
4) *Elimination of False Matching Point-Pairs:* In our experiments, we combine NNDR and GMS to obtain matching pairs with high accuracy.
5) *Image Transformation:* After obtaining candidate matching point pairs, RANSAC [34] is used to compute the image transformation matrix. It is supported by most candidate matching point-pairs. According to the estimated transformation matrix, the target image is transformed and aligned with the reference image.

In feature-based image registration, the feature extraction step plays a crucial role. The extracted features have a

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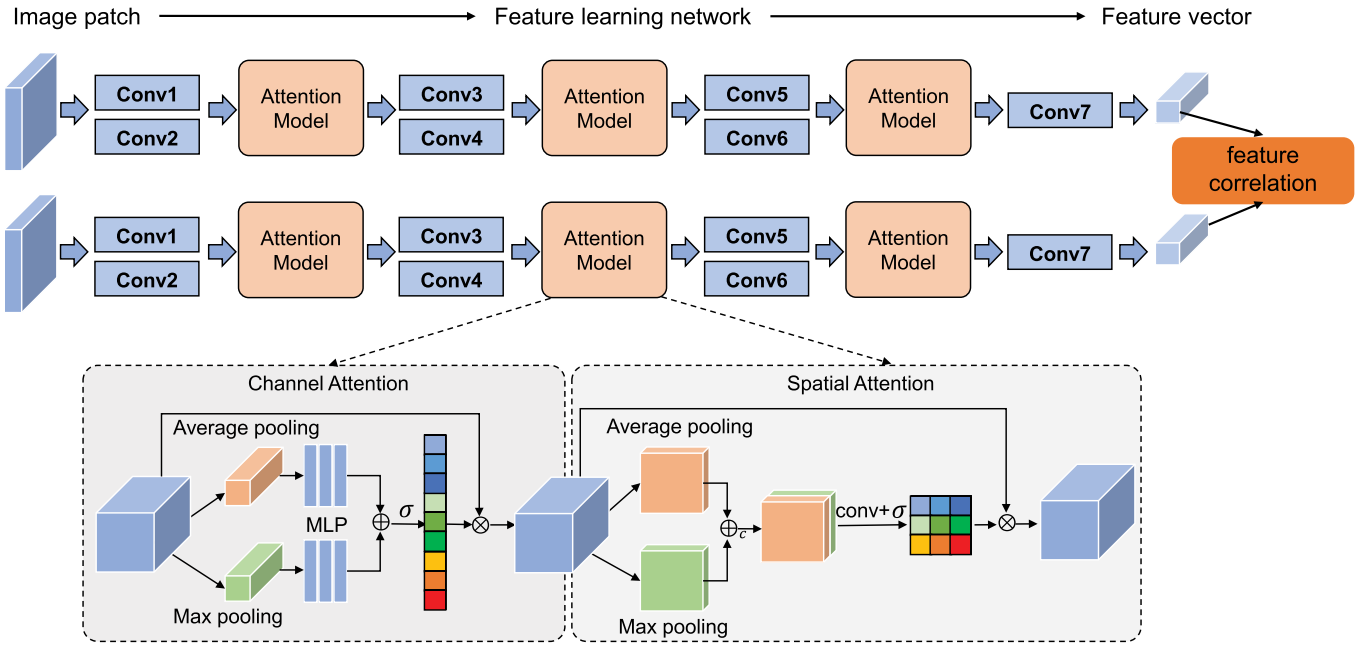

Fig. 1. Framework of the proposed deep feature Cnet.

great influence on image registration. Compared with handcrafted features, deep-learning-based features have significant advantages. Thus, we focus on improving the discrimination and representation of deep features and then enhancing the performance of image registration. Specifically, this article proposes a deep feature Cnet that improves feature discrimination from two perspectives. First, we build a deep feature learning network based on the attention mechanism, which encourages the model to pay more attention to meaningful features. Second, we design a feature correlation loss function to optimize the network, which ensures the feature learning network extracts more informative features and accelerates the network convergence. It should be noted that the proposed method is fully automatic, which is trained on a Liberty subset of the public image patch matching dataset [57]. Since the deep learning network has a strong generalization ability, the proposed deep feature Cnet that is trained on the natural local image patch training can perform well on multi-modal remote sensing image matching. In addition, the proposed method mainly focuses on dealing with image modality changes and the affine transformation (rotation transformation and translation transformation). In the following sections, we introduce our deep feature Cnet and its optimization method.

B. Network Structure

Taking the detected keypoints as the center, many local image patches are clipped and denoted as $x_i, i = 1, \ldots, N$, where N represents the number of local image patches. Then, we use a deep convolutional network to extract features from local image patches

$$f_i = F(x_i) \tag{7}$$

where f_i is the feature vector of the ith image patch, and F represents the function that maps image patches to feature vectors.

The general feature extraction network contains several convolutional blocks, convolutional layers, normalization layers, and nonlinear activation functions. In the process of feature learning, the feature at different positions is viewed equally. Actually, the features have different contributions for local feature matching. Concentrating on the informative features, we propose to build an attention feature learning network for remote sensing image registration based on the representative deep matching network, HardNet.

As illustrated in Fig. 1, our proposed feature learning network inserts the attention model into the convolutional layers. It contains seven convolutional layers and three AMs. Each convolutional layer is followed by a batch normalization layer and an activation function of Relu. The last convolutional layer does not have an activation function. Its output is flattened to a 128-dimensional feature vector. The parameters of the feature learning network are presented in Table I. As depicted in the bottom of Fig. 1, each AM learns the importance weights of the features along channel and spatial dimensions, which are labeled as channel attention and spatial attention, respectively.

Channel attention aims to learn the correlation and importance between channels of the extracted feature map. It first aggregates the information in a feature map $f^l \in R^{w \times h \times c}$ into a feature vector $f_d^l \in R^{1 \times 1 \times c}$ through the pooling operator. Following the attention learning module in CBAM [44], we use maxpooling (MP) and average pooling (AP) along the spatial dimension to acquire two feature vectors. Then, the multilayer perceptions (MLPs) are used to learn the initial channel attention weights, which are corresponding to different pooling operators. Finally, the attention weights are computed by fusing the initial channel attention weights of MP and AP.

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TABLE I
PARAMETERS OF FEATURE LEARNING NETWORK

Layers	Input size	Kernel size	Channel number	Stride	Output size
Conv1	32×32	3	32	1	32×32
Conv2	32×32	3	32	1	32×32
AM1	32×32	–	–	–	32×32
Conv3	32×32	3	64	2	16×16
Conv4	16×16	3	64	1	16×16
AM2	16×16	–	–	–	16×16
Conv5	16×16	3	128	2	8×8
Conv6	8×8	3	128	1	8×8
AM3	8×8	–	–	–	8×8
Conv7	8×8	8	128	1	1×1
Flatten	1×1	–	–	–	1×128

The learned channel attention feature map f_{ac}^{l} is formulated as

$$\begin{aligned} f_{\text{ac}}^{l} &= A_c(f^l) = f^l \otimes W_{\text{ac}} \\ W_{\text{ac}} &= [\sigma(\text{MLP}(\text{AP}_s(f^l)) + \text{MLP}(\text{MP}_s(f^l)))]_{\text{SR}} \end{aligned} \tag{8}$$

where A_c represents the channel attention learning operator, W_{ac} is the learned weight by channel AM, AP_s and MP_s are the AP and MP along the spatial dimensional, respectively, σ is the sigmoid function, $[\cdot]_{\text{SR}}$ represents the matrix replication along the spatial dimension and making the matrix size is $w \times h \times c$, and $\otimes$ is the Hadamard product.

Spatial attention aims to learn the importance weights of features at different spatial positions. It aggregates the information in a feature map $f^l \in R^{w \times h \times c}$ into a feature map $f_m^l \in R^{w \times h \times 1}$ through the pooling operator along the channel dimension. We also take different pooling methods of MP and AP to acquire two feature maps. Then, two feature maps are concatenated along the channel dimension. The spatial attention weights are computed by a convolutional layer and a sigmoid function. Finally, the input feature map multiplies spatial attention weights to acquire the spatial attention map

$$\begin{aligned} f_{\text{as}}^{l} &= A_s(f^l) = f^l \otimes W_{\text{as}} \\ W_{\text{as}} &= [\sigma(\text{conv}((\text{AP}_c(f^l)) \oplus_c (\text{MP}_c(f^l))))]_{\text{CR}} \end{aligned} \tag{9}$$

where A_s represents the spatial attention learning operator, W_{as} is the learned weight by spatial AM, AP_c and MP_c are the AP and MP along channel dimensional, respectively, conv is the convolutional layer, and $[\cdot]_{\text{CR}}$ represents the matrix replication along the channel dimension and making the matrix size is $w \times h \times c$.

To combine the channel attention and the spatial attention to form an attention model, we take the output of the channel attention learning model as the input of spatial learning model. The learned attention map can be written as

$$f_a^l = A_s(A_c(f^l)). \tag{10}$$

C. Network Optimization

The existing local feature learning networks usually map an image patch to an $L2$-normalized feature vector. Then, the similarity between features of patch pairs is measured by feature distance (e.g., Euclidean distance). Ideally, matching features should have a small feature distance and nonmatching features should have a large feature distance.

Contrastive loss focuses on the feature distance between single image patch pair. Given an image patch-pair (x_i, x_j), their corresponding descriptor vectors can be obtained through our proposed feature learning network

$$\begin{aligned} f_i &= F(x_i) \\ f_j &= F(x_j). \end{aligned} \tag{11}$$

The contrastive loss can be represented as

$$\begin{aligned} L_{\text{con},L2} &= \frac{1}{N_c} \sum_{i=1}^{N_c} y_{ij} d_{ij} + (1 - y_{ij})[m_{\text{dc}} - d_{ij}]_+ \\ d_{ij} &= D(f_i, f_j) = \left\| f_i - f_j \right\|_2 \end{aligned} \tag{12}$$

where N_c is the number of image patch pairs, d_{ij} is the feature distance between f_i and f_j, y_{ij} represents the matching label, if f_i and f_j are matching, $y_{ij} = 1$, else $y_{ij} = 0$, m_{dc} is the feature distance margin in the contrastive loss, and $[\cdot]_+$ means the minimum is 0.

As shown in (12), we can see that contrastive loss constrains the feature distance of matching image patch pairs to be as close as possible to zero, which is too strict for matching image patch pairs. However, due to significant differences caused by multiple sensors or geometric transformation, contrastive loss can easily result in overfitting in multi-modal image registration.

To solve this problem, triplet loss relaxes the constraint for matching image patch pairs and focuses on the relative distance between the matching image patch pair and the nonmatching image patch pair. Given a triplet $T_i = (x_{\text{ai}}, x_{\text{pi}}, x_{\text{ni}})$, x_{ai} and x_{pi} are matching, while x_{ai} and x_{ni} are nonmatching. Their corresponding descriptor vectors are f_{ai}, f_{pi}, and f_{ni}, respectively. The triplet loss is formulated as follows:

$$L_{\text{tri},L2} = \frac{1}{N_t} \sum_{i=1}^{N_t} \left[d_{\text{ap},i} - d_{\text{an},i} + m_{\text{dt}} \right]_+ \tag{13}$$

where N_t is the number of triplets, $d_{\text{ap},i}$ represents the feature distance of positive pair $(f_{\text{ai}}, f_{\text{pi}})$ in the ith triplet sample T_i, $d_{\text{an},i}$ represents the feature distance of negative pair $(f_{\text{ai}}, f_{\text{ni}})$ in T_i, and m_{dt} is a margin of feature distance in the triplet loss.

Unlike contrastive loss constraining the feature distance of matching patch pairs to be close to 0 and the feature distance of nonmatching patch pairs to be larger than margin m_{dc}, triplet loss expects the feature distance of negative patch pair to be larger than the feature distance of positive patch pair by a margin m_{dt}, that is, $d_{\text{an},i} - d_{\text{ap},i} \geq m_{\text{dt}}$.

To learn more discriminative features, we propose to use the feature correlation to measure feature similarity. It can be defined as

$$c_{ij} = f_i f_j^T. \tag{14}$$

The feature-correlation-based triplet loss function is as follows:

$$L_{\text{tri},c} = \frac{1}{B} \sum_{i=1}^{B} \left[c_{\text{an},i} - c_{\text{ap},i} + m_c \right]_+ \tag{15}$$

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where $c_{\mathrm{ap},i}$ represents the feature correlation of positive pairs in the ith triplet samples T_i, $c_{\mathrm{an},i}$ represents the feature correlation of negative pairs in T_i, and m_c is the margin of feature correlation. It expects the feature correlation of the positive patch pair to be larger than the feature correlation of the negative patch pair by a margin m_c, that is, $c_{\mathrm{ap},i} - c_{\mathrm{an},i} \geq m_c$.

However, the margin in the triplet loss function will significantly influence the image patch matching results and it is time-consuming to determine the optimal margin. Meanwhile, it is not unreasonable for triplet loss to use a fixed margin. As the learning ability of the deep feature extraction network improves, the margin should change dynamically from small to large. Thus, we propose a novel feature correlation loss to optimize the feature learning network

$$L_c = \frac{1}{N_t}\sum_{i=1}^{N_t}\log\left(1 + e^{s\left(c_{\mathrm{an},i} - c_{\mathrm{ap},i}\right)}\right) \tag{16}$$

where s represents the scale factor.

The gradients of the proposed feature correlation loss with respect to $c_{\mathrm{ap},i}$ and $c_{\mathrm{an},i}$ are derived as follows:

$$\frac{\partial L_c}{\partial c_{\mathrm{ap},i}} = \frac{-s e^{s\left(c_{\mathrm{an},i} - c_{\mathrm{ap},i}\right)}}{1 + e^{s\left(c_{\mathrm{an},i} - c_{\mathrm{ap},i}\right)}} \tag{17}$$

$$\frac{\partial L_c}{\partial c_{\mathrm{an},i}} = \frac{s e^{s\left(c_{\mathrm{an},i} - c_{\mathrm{ap},i}\right)}}{1 + e^{s\left(c_{\mathrm{an},i} - c_{\mathrm{ap},i}\right)}}. \tag{18}$$

According to the gradient in (17) and (18), the scale factor s can significantly enlarge the gradient amplitude and accelerate the convergence of the feature learning network.

In summary, the proposed feature correlation loss L_c expects the matching patches to have high correlation and the nonmatching patches to have lower correlation. It encourages the network to extract the invariant features for matching patches and learn more discriminative features for nonmatching patches. Meanwhile, L_c does not need to search the optimal margin parameters. Moreover, our proposed feature correlation loss with the scale factor s can further enhance the feature learning and accelerate the network convergence. The network optimization is illustrated in Algorithm 1.

Algorithm 1 Training Feature Learning Network on One Minibatch

Input: Generated triplet image patches $\{T_i = (x_{ai}, x_{pi}, x_{ni}), i = 1 \ldots N_t\}$, batch size B, learning rate β, scale factor s, initial feature extraction function F, its network parameters are represented as θ.

Output: Updated feature learning function F.

1: **Step 1:** Feedforward triplet samples into feature extraction network to obtain descriptor of each image patch, f_{ai}, f_{pi}, and f_{ni}.
2: **Step 2:** Compute feature correlation of matching patch pair $c_{ap,i}$ and nonmatching patch pair $c_{an,i}$, according to14.
3: **Step 3:** Compute feature correlation loss functions L_c based on16.
4: **Step 4:** Compute gradients and update parameters:
5: $\theta = \theta - \beta\frac{\partial L_c}{\partial \theta}$.
6: **return** θ.

IV. Experiments

Local feature matching is a critical step in image registration and its improvement of this article. To show the effectiveness of our proposed method more intuitively, we first test it on the local image patches' matching datasets of Brown [57] and HPatches [58]. After that, we evaluate the image registration performance on various datasets, including cross-spectral image registration (VIS-NIR), multi-modal (heterogeneous) remote sensing image registration, and single-modal (homogeneous) remote sensing image registration. The following sections introduce the datasets and implementation details.

A. Implementation Details

The network was optimized by a stochastic gradient descent (SGD) optimizer with momentum 0.9, weight decay 10^{-4} and initial learning rate 10. The batch size is 1024. We performed data augmentation through random flip and rotation to enhance the network robustness. In the training process, we adopt the variance constraint to minimize the correlation variance of matching and nonmatching samples. We also adopt the hard negative samples' mining strategy for network optimization [20]. The experiments were conducted using an NVIDIA GTX 2080 Ti. In image registration experiments, the number of keypoints is 20 000.

B. Evaluation Metrics

We adopt the false positive rate value at the true positive rate equal to 95% (FPR95) to test the matching results, which can reflect the performances of the local descriptor learning methods. The smaller the FPR95, the better the matching performance.

To evaluate the performance of patch verification, image matching, and patch retrieval on the HPatches dataset, we present the mean average precision (mAP) on these three tasks, with a higher mAP, indicating a higher performance.

Projection error is a common evaluation criterion in image registration, which is defined as the root mean square error (RMSE) of all matching keypoints

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{M}\left(\left(lx_i - lx_i{}'\right)^2 + \left(ly_i - ly_i{}'\right)^2\right)}{M}} \tag{19}$$

where lx_i and ly_i are the locations of the ith matching points, lx_i' and ly_i' are the predicted positions of the ith matching points, and M is the number of acquired matching points. RMSE is normalized by the pixel size [59]. To show the performance of image registration in more detail, we present the distribution of RMSE, such as the number of images with RMSE less than 1, the number of images with RMSE between 1 and 3, and so on. On the other hand, the number of true matching point pairs (Matches) and the number of register images (Registers) also reflect the performance of image registration.

It should be noted that RMSE just reflects coherent matching pairs. It cannot be used to test the image registration

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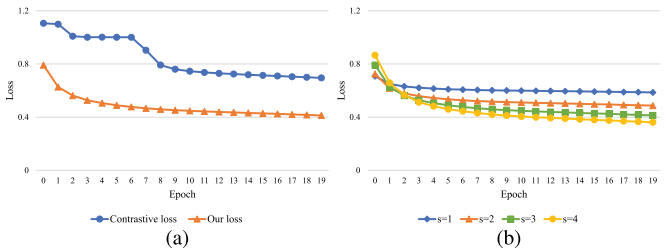

Fig. 2. (a) Loss changing curves based on contrastive loss and our proposed loss. (b) Loss changing curves based on our loss with different scale factors.

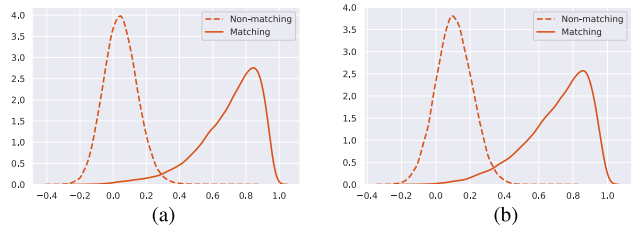

Fig. 3. Feature correlation distributions of matching and nonmatching image patch pairs on the Notredame and Yosemite datasets. (a) Notredame. (b) Yosemite.

performance alone [59]. We should combine RMSE and some visual results to present the image registration performances, such as the matching points' connection images, checkerboard images, and register visualization result.

C. Image Patch Matching on Brown Dataset

This section mainly presents the effectiveness of our proposed method on the public image patch dataset, Brown.

1) Brown Dataset: Brown dataset is widely used for local descriptor evaluation [57]. It has three subsets, including Liberty, Notredame, and Yosemite. There are 100K, 200K, and 500K labeled matching and nonmatching image patch pairs, respectively. The matching image patch pairs have illumination change, rotation transformation, and perspective changes. Following previous studies [19], [20], [38], [39], we train the deep learning network on one subset and test on the other two subsets.

2) Brown Results: Table II shows the matching results of Cnet with AM and without AM based on different loss functions. First, Cnet with AM achieves better mean FPR95 results than Cnet without AM based on contrastive loss, ratio loss, and feature correlation loss. Second, we can see that the proposed feature correlation loss has significant advantages over other losses. Specifically, when Cnet does not contain AM, the feature correlation loss improves the mean FPR95 values of other losses by 2.19, 0.48, and 0.24, respectively. When Cnet includes AM, the feature correlation loss improves the mean FPR95 values of other losses by 1.94, 0.57, and 0.55, respectively. These results show the effectiveness of the proposed network structure and loss function.

In addition, it should be noted that the contrastive loss has the worst matching results between these optimization loss functions. The main reason is that contrastive loss constrains the feature distance of matching samples to be as small as possible, which is too strict for matching samples, especially on multi-modal samples with image modality changes. Thus, contrastive loss will lead to an unstable training process and increase the risk of overfitting. As shown in Fig. 2(a), we can see that the training process based on contrastive loss is unstable. In contrast, our proposed feature correction loss relaxes the constraint on matching multi-modal images, which focuses on the relative feature similarity between matching and nonmatching samples. Thus, its training process is very stable. Moreover, we introduce a scale factor s in feature correction, which can accelerate network training and convergence. As shown in Fig. 2(b), we present the loss change curves trained on the Liberty subset. We can see that the bigger the scale factor s, the faster the network convergence.

In the training process of the deep matching network, the batch size and learning rate will influence the matching results. To find the optimal parameters, we test the FPR95 results based on different batch sizes (128, 256, 512, and 1024) and learning rates (0.1, 1.0, and 10). As shown in Table III, the matching results are increased gradually with the increase in batch size. The main reason is that we adopt the hard negative samples mining between the batch samples for network optimization. The large batch size will help the network select the representative hard negative samples, further enhancing network optimization. Considering the memory limitation and the matching performance, we set the batch size as 1024 in our experiments. As shown in Table IV, the bigger learning rate will achieve better matching results. The matching results are more sensitive to the learning rate than the batch size. Generally, the network will fall into local optimization if the learning rate is too small. In contrast, the network optimization is unstable if the learning rate is too large. Following the same experimental setting as HardNet, we set the learning rate as 10.

To further show the effectiveness of the feature Cnet on image matching, we display the feature correlation distributions of matching and nonmatching image patch pairs by the kernel density estimation (KDE) method. As shown in Fig. 3, the matching patch pairs have a larger feature correlation than nonmatching patch pairs on the Notredame and Yosemite datasets. Meanwhile, there is a significant margin between the feature correlation distributions of matching and nonmatching image patch pairs. These results demonstrate the effectiveness of our method, which can extract the invariant features for matching image patch pairs with various changes and discriminative features for identifying nonmatching image patch pairs.

D. Image Patch Matching on HPatches Dataset

To demonstrate the effectiveness of our proposed method, we evaluated it on the public image patch dataset, HPatches.

1) HPatches Dataset: The HPatches dataset [58] is a popular benchmark for evaluating local image descriptors. It contains 59 image sequences with obvious perspective transformation and 57 image sequences with illumination transformation. Each of image sequence contains one reference image and five target images. According to the difficulty, each sequence can be divided into easy, hard, and tough. Three tasks

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TABLE II
FPR95 RESULTS OF DIFFERENT NETWORK STRUCTURES AND LOSS FUNCTIONS ON THE BROWN DATASET

Training	Liberty		Notredame		Yosemite		
Test	Notredame	Yosemite	Liberty	Yosemite	Liberty	Notredame	Average
Cnet-Contrast	1.15	4.90	2.61	4.34	5.57	1.78	3.39
Cnet-Ratio	0.64	2.41	1.55	1.69	2.86	0.90	1.68
Cnet-Triplet	0.53	1.84	1.49	1.51	2.51	0.78	1.44
Cnet-Our loss	0.40	1.40	1.20	1.28	2.28	0.62	1.20
Cnet+AM-Contrast	1.01	3.87	2.47	3.06	5.61	1.80	2.97
Cnet+AM-Ratio	0.59	2.11	1.39	1.68	2.83	0.98	1.60
Cnet+AM-Triplet	0.66	2.04	1.47	1.43	2.95	0.93	1.58
Cnet+AM-Our loss	**0.36**	**1.24**	**1.05**	**0.99**	**1.86**	**0.70**	**1.03**

TABLE III
FPR95 RESULTS OF THE BROWN DATASET BASED ON DIFFERENT BATCH SIZES

Training	Liberty		Notredame		Yosemite		
Test	Notredame	Yosemite	Liberty	Yosemite	Liberty	Notredame	Average
128	0.54	1.78	1.58	2.17	2.99	1.01	1.68
256	0.47	1.66	1.39	1.42	2.74	1.04	1.45
512	0.45	1.64	1.28	1.08	2.57	0.85	1.31
1024	**0.36**	**1.24**	**1.05**	**0.99**	**1.86**	**0.70**	**1.03**

TABLE IV
FPR95 RESULTS OF THE BROWN DATASET BASED ON DIFFERENT LEARNING RATES

Training	Liberty		Notredame		Yosemite		
Test	Notredame	Yosemite	Liberty	Yosemite	Liberty	Notredame	Average
0.1	0.78	2.75	2.81	2.34	4.32	1.44	2.41
1.0	0.47	1.57	1.44	1.34	2.48	0.77	1.35
10.0	**0.36**	**1.24**	**1.05**	**0.99**	**1.86**	**0.70**	**1.03**

are used to evaluate descriptor performance: patch verification, image matching, and patch retrieval. Following the evaluation criteria from previous articles [38], [58], our feature learning network is also trained on the Liberty subset of the Brown dataset [57].

2) HPatches Results: Table V lists the mean results of patch verification, image matching, and patch retrieval based on different scales s. It can be seen that the optimal scale s in (16) is 3.0. As the scale s increases to 4.0, the performance drops slightly. In other experiments, therefore we set $s = 3.0$.

In addition, we compare our proposed method with some representative local feature descriptor methods, including HardNet [20], SOSNet [36], and Softmargin [37]. HardNet is an important baseline in local feature learning. Thus, we build our Cnet based on the feature network structure of HardNet and use our proposed feature correlation loss to optimize Cnet. In addition, our proposed attention feature learning network with feature correlation loss is denoted as Cnet + AM.

Fig. 4 shows the results of patch verification, image matching, and patch retrieval on HPatches. Although the performance of our proposal is slightly lower than those of Softmargin on the verification task, our method has significant advantages on image matching and patch retrieval. Compared with HardNet [20], SOSNet [36], and Softmargin [37], the performance of our method on image patch matching is higher by 1.65%, 2.97%, and 1.16%, respectively, and the performance on image patch retrieval is higher 1.87%, 1.23%, and 1.64%, respectively. These experimental results demonstrate the effectiveness of our Cnet.

TABLE V
MEAN ACCURACY RESULTS BASED ON VARIOUS SCALE s ON THE HPATCHES DATASET

s	Patch Verification	Image Matching	Patch Retrieval
1.0	88.25%	54.38%	71.02%
2.0	88.48%	53.93%	70.94%
3.0	**88.66%**	**54.41%**	**71.53%**
4.0	88.42%	53.78%	70.87%

TABLE VI
IMAGE REGISTRATION RESULTS OF CROSS-SPECTRAL IMAGES VIS-NIR. *INDICATE WITHOUT GMS FOR ELIMINATING FALSE MATCHING POINT PAIRS

Method	Registers	RMSE distribution			Matches
		$0-1$	$1-3$	$3-10$	
SIFT*	25	17	3	5	1401
SIFT	34	20	8	6	782
ORB*	25	10	8	7	2243
ORB	45	25	12	8	823
HardNet	46	18	20	8	1363
SOSNet	46	22	15	9	1417
Softmargin	45	20	17	8	1565
Cnet	46	19	18	9	1451
Cnet+AM	46	20	18	8	1874

E. Cross-Spectral Image Registration

This section presents the results of our proposed method on cross-spectral image registration: VIS-NIR dataset. First, ORB is used to detect keypoints. Second, the feature learning network trained on Liberty is used for local description extraction. Third, we use the NNDR matching strategy to obtain initial

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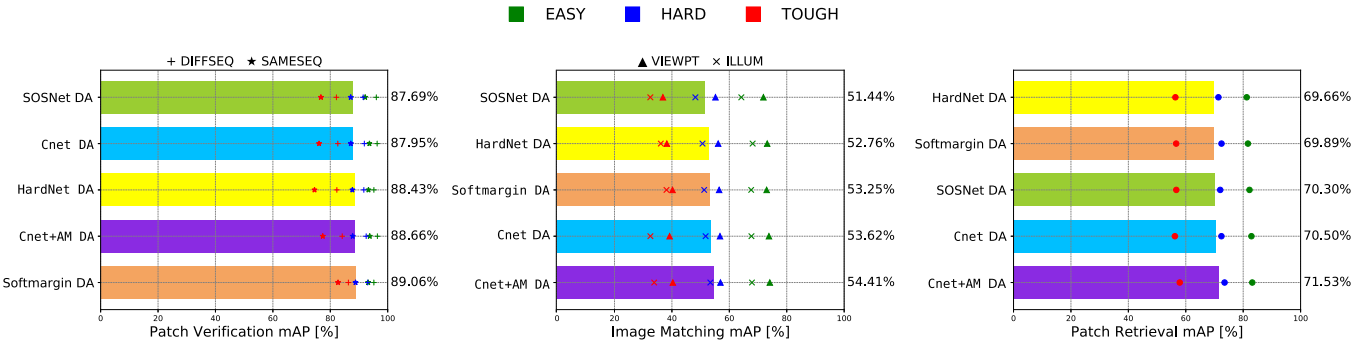

Fig. 4. Comparison mean accuracy results on image patches' dataset of HPatches.

Fig. 5. Two pairs of visible (VIS) and near-infrared (NIR) images were sampled from the field subset in VIS-NIR. (a) VIS image-1. (b) NIR image-1. (c) VIS image-2. (d) NIR image-2.

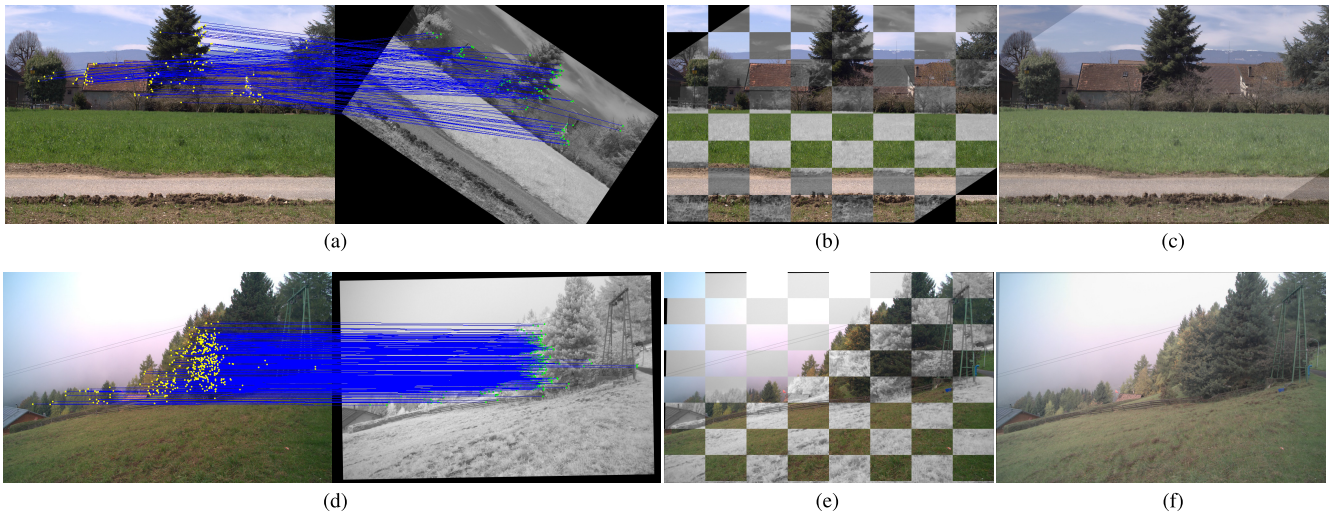

Fig. 6. Visual registration results of the cross-spectral VIS-NIR images. (a) Matching points. (b) Checkerboard. (c) Registered image. (d) Matching points. (e) Checkerboard. (f) Registered image.

TABLE VII

DETAILS OF THE MULTI-MODAL REMOTE SENSING IMAGES. "GSD" MEANS GROUND SAMPLE DISTANCE

Image	Modal	Size	GSD	Data
Multi-modal-a	Airborne visible	480 × 480	2.5m	06/2012
	LiDAR depth	480 × 480	2.5m	06/2012
Multi-modal-b	Google Earth	510 × 510	3.0m	11/2007
	TerraSAR-X	510 × 510	3.0m	12/2007
Multi-modal-c	WorldView2 visibel	570 × 600	2.0m	10/2011
	LiDAR intensity	570 × 600	2.0m	10/2010
Multi-modal-d	Landsat 5 TM band3	600 × 600	30m	05/2007
	TerraSAR-X	600 × 600	30m	03/2008

matching point pairs. Then, GMS and RANSAC are used to eliminate false matching point pairs and estimate the image transformation parameters.

1) VIS-NIR Images: VIS-NIR is a public cross-spectral dataset [60]–[62], consisting of many pairs of visible and near-infrared images. It can be acquired from online [63]. VIS-NIR is usually used to evaluate the performance of descriptor learning on multi-modal images. It has nine subsets: country, field, forest, indoor, mountain, old building, street, urban, and water. We perform image registration tests on the challenging field subset. Filed subset contains 51 pairs of VIS-NIR images. From Fig. 5, we can see that there are many repetitive local patterns and weak textures in the images of the Field subset. In addition to cross-spectral differences, we also conduct affine transformations on NIR images, including random scale changes ± 0.2 and rotation transformation ± 35 °s.

2) VIS-NIR Images Registration Results: Table VI shows the cross-spectral image registration performance of various methods, such as SIFT [11], ORB [24], HardNet [20], SOSNet [36], Softmargin [37], and our Cnet. Cnet is capable of acquiring many matching points and registering 46 pairs of cross-spectral images (in total 51 image pairs). Table VI also gives the statistical histogram distribution of RMSE values,

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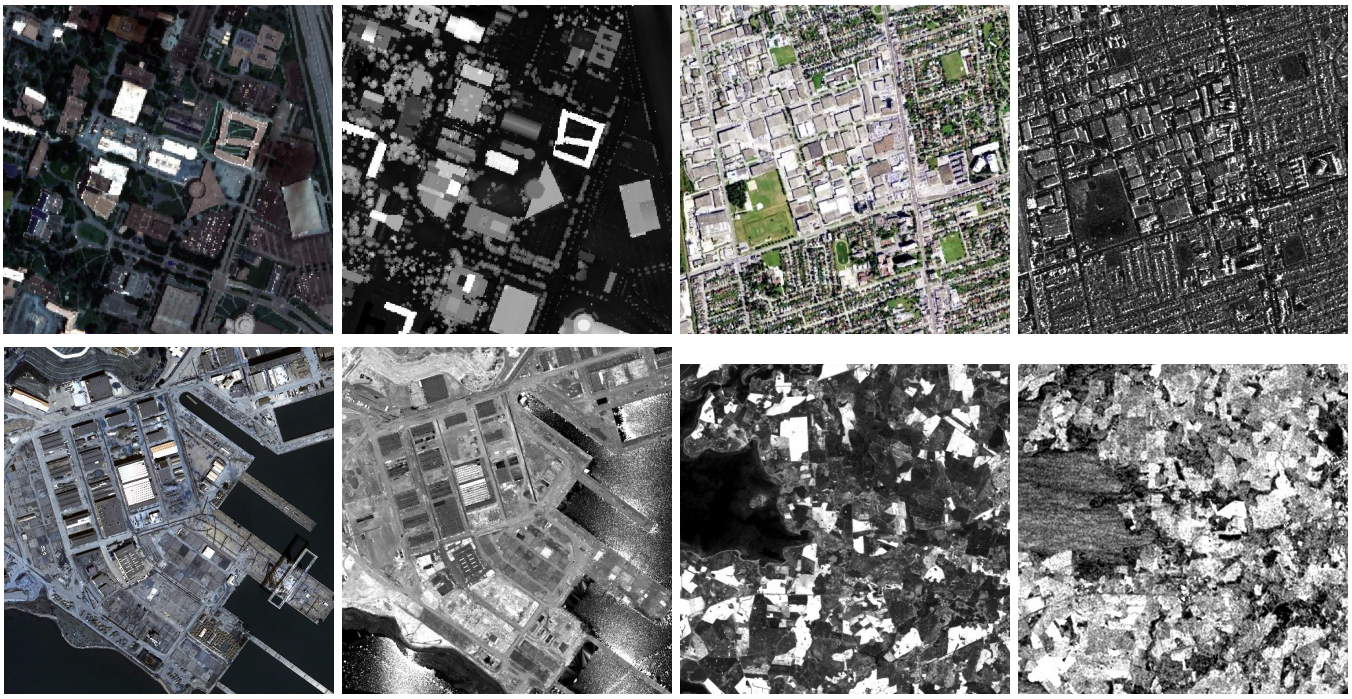

Fig. 7. Four pairs of multi-modal remote sensing images, which are multi-modal-a, multi-modal-b, multi-modal-c, and multi-modal-d, respectively.

such as the number of images with RMSE less than 1, the number of images with RMSE between 1 and 3, and the number of images with RMSE between 3 and 10. It can be seen that most images achieve the subpixel error (RMSE $\leq$ 1). Fig. 6 presents the visualization results of two pairs of images. It shows that our proposed method can provide accurate image registration on cross-spectral images.

F. Multi-Modal Remote Sensing Image Registration

To verify the effectiveness of our proposed method, we test it on four pairs of multi-modal remote sensing image.

1) Multi-Modal Remote Sensing Images: The multi-modal remote sensing image pairs contain visible-LiDAR image and visible-SAR image. More details about multi-modal images are presented in Table VII. As shown in Fig. 7, there are great differences between multi-modal images and significant noise in SAR images. Therefore, it is difficult to accurately register these multi-modal remote sensing images. Following the same settings as [4], [5], [15], there are slight translation changes in test multi-modal remote sensing images. These multi-modal images have been preregistered using metadata or physical sensor models and resampled to the same resolution [64], [65]. The main challenge is to deal with the significant differences between multi-modal images caused by different imaging mechanisms.

2) Multi-Modal Remote Sensing Image Registration Results: Fig. 8 shows the visualization results of image registration. It shows that our method obtains many matching point pairs and achieves a high registration performance. The quantitative results in Table VIII show that traditional handcrafted features cannot register these multi-modal images, but deep descriptor learning methods perform well on all multi-modal remote sensing images. These results demonstrate that deep-learning-based methods are more suitable for multi-modal image registration and are generalizable from nature images to remote sensing images. Except for multi-modal-a, our method achieves the highest performances and finds more corresponding point pairs than the other methods on the multi-modal remote sensing images. Especially in multi-modal-d, the serious level of noise introduces more difficulties for image registration. However, our method improves the registration performance by 22.22%–40.17% in multi-modal-d, which provides compelling evidence for the robustness of our proposal. Compared with the baseline HardNet, our method performs better on all the multi-modal remote sensing images. So, our proposed feature correlation loss function L_c outperforms feature-distance-based triplet losses. This is because our proposed feature correlation loss not only enables the network to extract the common features for matching samples but also encourages the network to learn more discriminative features for nonmatching samples. Moreover, the registration results of Cnet + AM surpass those of Cnet, which shows the effectiveness of our proposed feature learning network based on the attention mechanism.

G. Yellow River Remote Sensing Image Registration

To further show the generalization of our proposal, we also register three pairs of remote sensing image about Yellow River Estuary in China.

1) Yellow River Remote Sensing Images: As shown in Fig. 9, the SAR images were acquired by the Radarsat-2 in June 2008 and June 2009. The first and the second image pairs were with the size of 1000 $\times$ 1000. The size of the third image pair is 921 $\times$ 593, and the corresponding optical image is acquired from Google Earth. There

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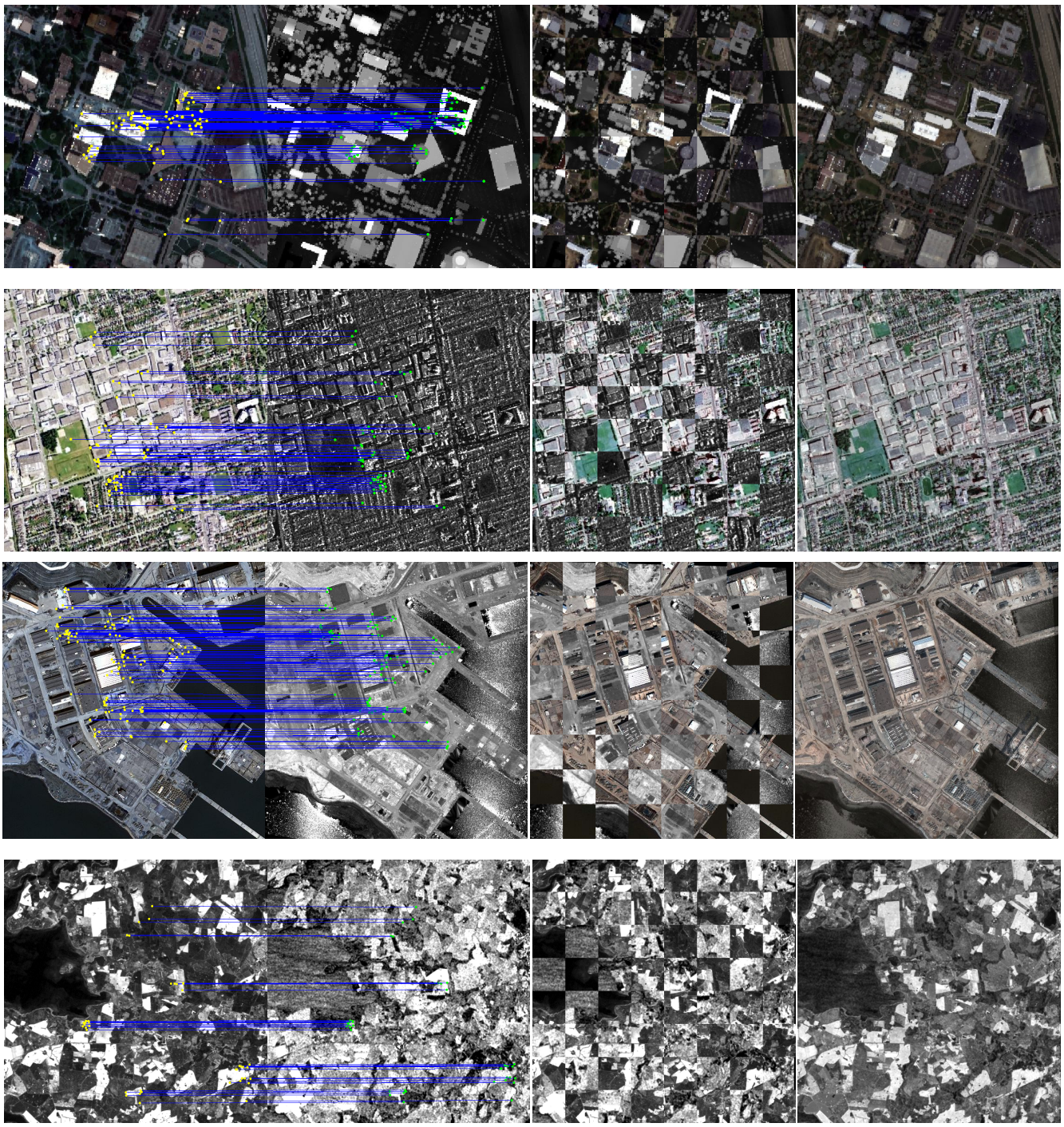

Fig. 8. Visual registration results of multi-modal remote sensing images. (From Left to Right) Matching points, checkerboard images, and registered images.

are slight rotation transformations, small translations, and image modality changes in these Yellow River remote sensing images.

2) Yellow River Remote Sensing Image Registration Results: The visualization results are shown in Fig. 10, and the quantitative results are presented in Table IX. It can be seen that the lines and textures in the checkerboard images are continuous and smooth, which indicates a high registration performance. The quantitative results show that our proposed method achieves the subpixel error on all Yellow River images. Specifically, the results of RMSE on Yellow River images are 0.40, 0.49, and 0.61, respectively. Extensive experiments demonstrate that our proposed method is effective and robust on various remote sensing image registration tasks.

H. Image Registration Time

Moreover, we present the time details of image registration in Table X, including keypoint detection, feature extraction, feature matching, false matching point pairs' elimination, image register, and the total time. The average time of image

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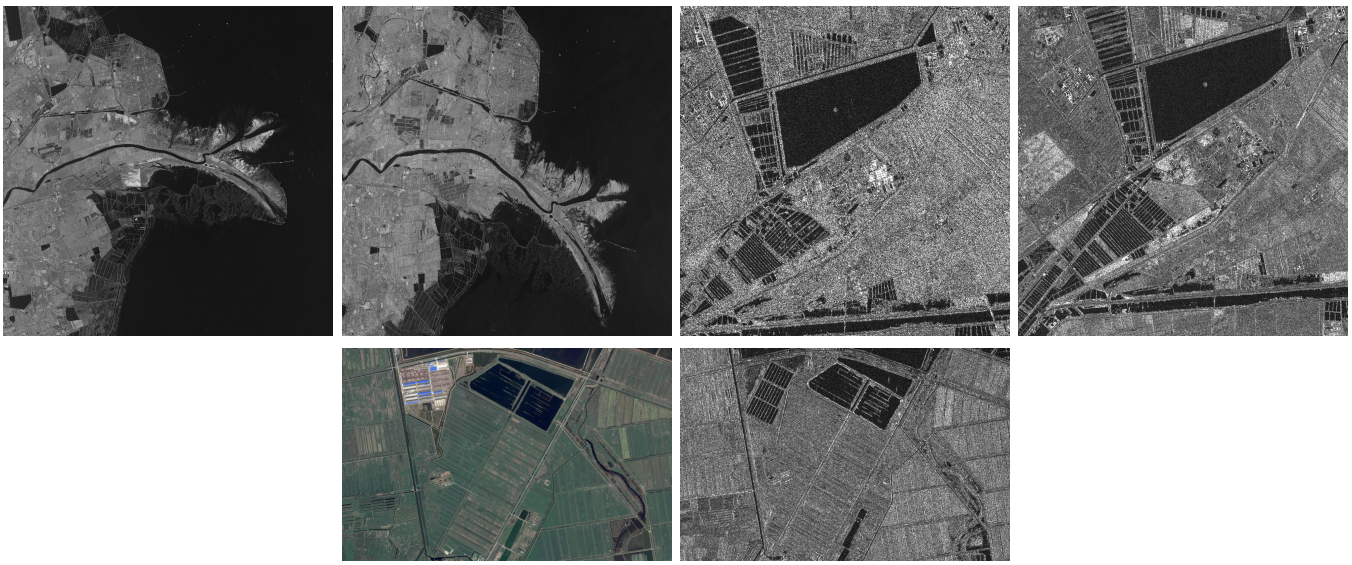

Fig. 9. Three pairs of remote sensing images about Yellow River, which are labeled as YellowRiver-a, YellowRiver-b, and YellowRiver-c, respectively.

TABLE VIII

REGISTRATION RESULTS OF MULTI-MODAL REMOTE SENSING IMAGES. BOLDFACE INDICATES THE BEST RESULT

Method	Multi-modal-a		Multi-modal-b		Multi-modal-c		Multi-modal-d		Multi-modal-mean	
	RMSE	Matches	RMSE	Matches	RMSE	Matches	RMSE	Matches	RMSE	Matches
SIFT	-	-	-	-	-	-	-	-	-	-
ORB	-	-	-	-	-	-	-	-	-	-
HardNet-Contrast	0.68	111	0.72	56	0.76	152	**0.61**	39	0.69	90
HardNet-Ratio	**0.48**	170	0.59	84	0.76	167	0.67	34	0.63	114
HardNet	0.68	104	0.72	98	0.66	161	0.90	77	0.74	110
SOSNet	**0.48**	135	0.59	102	0.67	134	0.97	108	0.68	120
Softmargin	0.67	107	0.94	99	0.74	175	1.17	73	0.88	114
D2net	0.89	65	0.94	71	0.75	70	0.96	45	0.89	63
Cnet	0.65	95	0.66	78	0.63	156	0.84	77	0.70	102
Cnet+AM	0.58	108	**0.58**	134	**0.61**	253	0.70	104	**0.62**	150

TABLE IX

REGISTRATION RESULTS OF YELLOW RIVER REMOTE SENSING IMAGES

Method	YellowRiver-a		YellowRiver-b		YellowRiver-c		YellowRiver-mean	
	RMSE	Matches	RMSE	Matches	RMSE	Matches	RMSE	Matches
SIFT	-	-	-	-	-	-		
ORB	-	-	-	-	-	-		
HardNet-Contrast	0.75	183	0.74	154	0.67	63	0.72	133
HardNet-Ratio	0.74	269	0.78	254	0.67	78	0.73	200
HardNet	0.50	367	0.68	262	0.73	208	0.64	279
SOSNet	0.41	416	0.98	100	0.95	60	0.78	192
Softmargin	0.47	316	**0.47**	228	0.68	125	0.54	223
D2net	0.45	63	**0.47**	259	0.72	188	0.55	170
Cnet	0.53	333	0.50	190	0.87	153	0.63	225
Cnet+AM	**0.40**	303	0.49	250	**0.61**	160	**0.50**	238

TABLE X

COMPUTATION TIME (S) OF IMAGE REGISTRATION

Images	Keypoint detection	Feature extraction	Matching	Elimination	Register	Total
Multi-modal-a	0.0881	0.7833	0.5113	0.2136	0.0221	1.6184
Multi-modal-b	0.1027	0.8622	0.6897	0.2396	0.0324	1.9266
Multi-modal-c	0.1230	0.8899	0.7695	0.2419	0.0462	2.0705
Multi-modal-d	0.1252	0.9309	0.7383	0.2734	0.0265	2.0942
YellowRiver-a	0.1989	0.9404	0.8979	0.2290	0.0828	2.3490
YellowRiver-b	0.2209	0.9452	0.8161	0.2902	0.0466	2.3190
YellowRiver-c	0.1507	0.9406	0.8763	0.2461	0.0308	2.2444
Average time	0.1442	0.8989	0.7570	0.2477	0.0411	2.0889

registration on seven remote sensing images is 2.0889 s. It can be seen that feature extraction and feature matching take up a lot of time because there are 20 000 ORB keypoints in each remote sensing image for matching. Thus, we will study fast image registration methods under the precondition of high image registration accuracy through reducing the number of keypoints and increasing the speed of feature extraction and feature matching.

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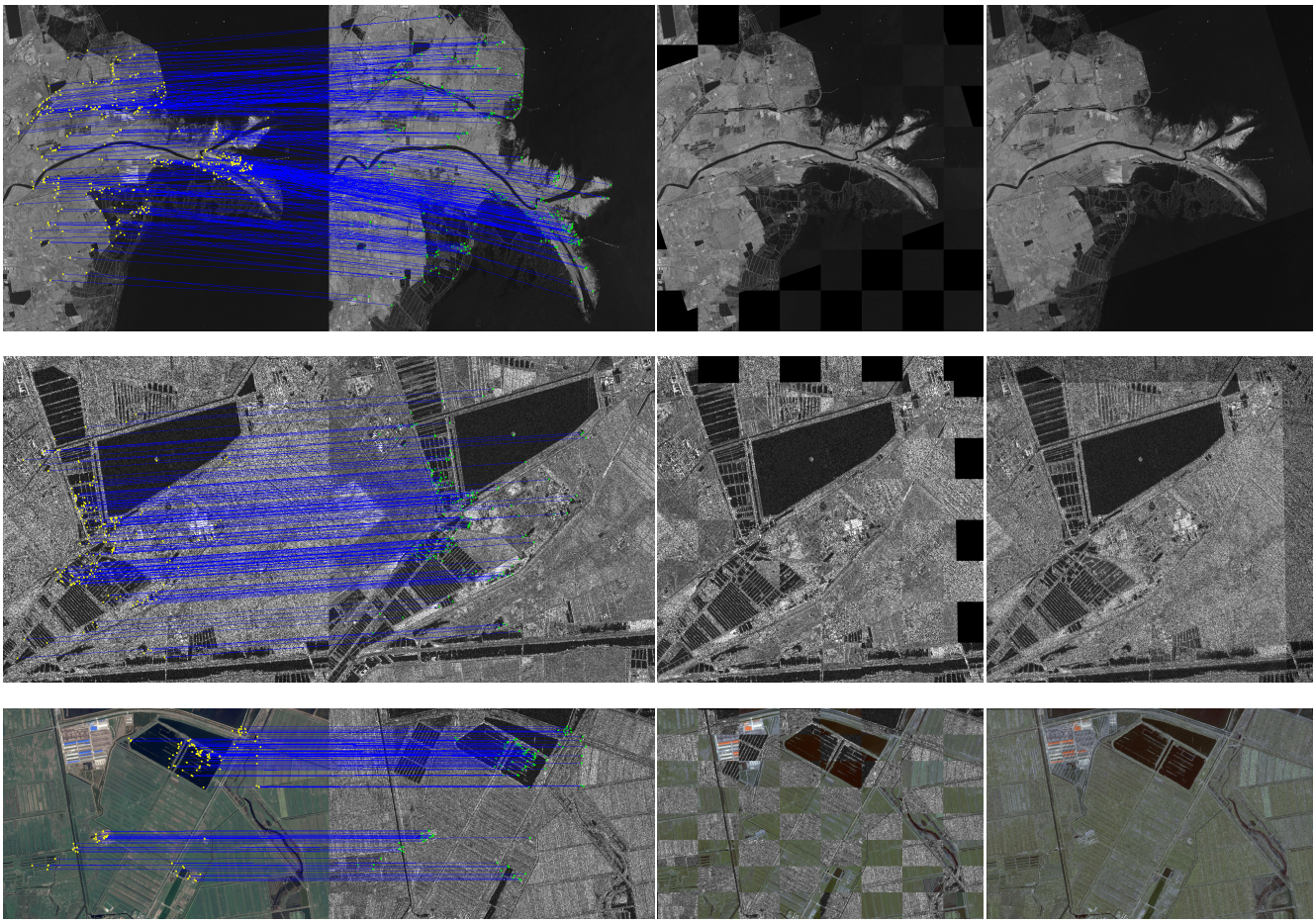

Fig. 10. Visual registration results of Yellow River images. (From Left to Right) Matching points, checkerboard images, and registered images.

V. Conclusion

This article proposes a deep feature Cnet for multi-modal remote sensing image registration. Cnet builds a feature learning network based on the attention mechanism. It forces the model to pay more attention to meaningful features and improves the image registration performance. Moreover, we design a novel feature correlation loss for network optimization. It can extract the invariant features for identifying the matching image patches with illumination changes, rotation transformations, and image modality changes. Meanwhile, it can learn the discriminative features for separating the nonmatching image patches. In addition, the proposed loss can accelerate network convergence and enhance network stability in the training process. Extensive experimental results on local image patch matching, cross-spectral image registration, multimodal remote sensing image registration, and sing-modal remote sensing image registration have demonstrated the effectiveness and robustness of our proposed method.

Although deep-learning-based methods acquired better performances than handcrafted features on image matching/registration, they contain several hyperparameters that will affect the optimization speed and performances of the network. Therefore, we usually need to rely on engineering experience or incur significant time cost to find the optimal hyperparameter settings. In the follow-up research, we will study automatic identification of optimal hyperparameters or reduce the sensitivity of matching results to these parameters and improve the stability of the deep matching network. Meanwhile, we will focus on improving the speed of image registration. We also want to use Cnet on MindSpore, which is a new deep learning computing framework.

References

[1] M. Gong, L. Su, M. Jia, and W. Chen, "Fuzzy clustering with a modified MRF energy function for change detection in synthetic aperture radar images," *IEEE Trans. Fuzzy Syst.*, vol. 22, no. 1, pp. 98–109, Feb. 2014.

[2] Y. Xing, S. Yang, Z. Feng, and L. Jiao, "Dual-collaborative fusion model for multispectral and panchromatic image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[3] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[4] Y. Ye and J. Shan, "A local descriptor based registration method for multispectral remote sensing images with non-linear intensity differences," *ISPRS J. Photogramm. Remote Sens.*, vol. 90, pp. 83–95, Apr. 2014.

[5] Y. Ye, J. Shan, L. Bruzzone, and L. Shen, "Robust registration of multimodal remote sensing images based on structural similarity," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2941–2958, Mar. 2017.

[6] K. Ravani, S. Saboo, and J. S. Bhatt, "A practical approach for SAR image despeckling using deep learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 2957–2960.

[7] F. Maes, A. Collignon, D. Vandermeulen, G. Marchal, and P. Suetens, "Multimodality image registration by maximization of mutual information," *IEEE Trans. Med. Imag.*, vol. 16, no. 2, pp. 187–198, Apr. 1997.

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[8] J. Liang, X. Liu, K. Huang, X. Li, D. Wang, and X. Wang, "Automatic registration of multisensor images using an integrated spatial and mutual information (SMI) metric," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 603–615, Jan. 2014.

[9] R. Feng, H. Shen, J. Bai, and X. Li, "Advances and opportunities in remote sensing image geometric registration: A systematic review of state-of-the-art approaches and future research directions," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 120–142, Dec. 2021.

[10] B. S. Reddy and B. N. Chatterji, "An FFT-based technique for translation, rotation, and scale-invariant image registration," *IEEE Trans. Image Process.*, vol. 5, no. 8, pp. 1266–1271, Aug. 1996.

[11] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, 2004.

[12] H. Bay, A. Ess, T. Tuytelaars, and L. Van Gool, "Speeded-up robust features (SURF)," *Comput. Vis. Image Understand.*, vol. 110, no. 3, pp. 346–359, Jan. 2008.

[13] G. Yu and J. M. Morel, "ASIFT: An algorithm for fully affine invariant comparison," *Image Process. On Line*, vol. 1, nos. 1232–2105, pp. 11–38, 2011.

[14] Y. Xiang, F. Wang, and H. You, "OS-SIFT: A robust SIFT-like algorithm for high-resolution optical-to-SAR image registration in suburban areas," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3078–3090, Jun. 2018.

[15] Y. Ye, L. Shen, M. Hao, J. Wang, and Z. Xu, "Robust optical-to-SAR image matching based on shape properties," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 4, pp. 564–568, Apr. 2017.

[16] X. Han, T. Leung, Y. Jia, R. Sukthankar, and A. C. Berg, "MatchNet: Unifying feature and metric learning for patch-based matching," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3279–3286.

[17] E. Simo-Serra, E. Trulls, L. Ferraz, I. Kokkinos, P. Fua, and F. Moreno-Noguer, "Discriminative learning of deep convolutional feature point descriptors," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 118–126.

[18] S. Zagoruyko and N. Komodakis, "Learning to compare image patches via convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 4353–4361.

[19] Y. Tian, B. Fan, and F. Wu, "L2-Net: Deep learning of discriminative patch descriptor in Euclidean space," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6128–6136.

[20] A. Mishchuk, D. Mishkin, F. Radenovic, and J. Matas, "Working hard to know your neighbor's margins: Local descriptor learning loss," 2017, *arXiv:1705.10872*.

[21] X. Zhang, F. X. Yu, S. Kumar, and S.-F. Chang, "Learning spread-out local feature descriptors," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4605–4613.

[22] J. S. Bhatt and N. Padmanabhan, "Image registration for meteorological applications: Development of a generalized software for sensor data registration at ISRO," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 23–37, Dec. 2020.

[23] J. N. Sarvaiya, S. Patnaik, and S. Bombaywala, "Image registration by template matching using normalized cross-correlation," in *Proc. Int. Conf. Adv. Comput., Control, Telecommun. Technol.*, Dec. 2009, pp. 819–822.

[24] E. Rublee, V. Rabaud, K. Konolige, and G. Bradski, "ORB: An efficient alternative to SIFT or SURF," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 2564–2571.

[25] S. Suri, P. Schwind, J. Uhl, and P. Reinartz, "Modifications in the SIFT operator for effective SAR image matching," *Int. J. Image Data Fusion*, vol. 1, no. 3, pp. 243–256, Sep. 2010.

[26] S. Wang, H. You, and K. Fu, "Bfsift: A novel method to find feature matches for SAR image registration," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 4, pp. 649–653, Jul. 2012.

[27] F. Dellinger, J. Delon, Y. Gousseau, J. Michel, and F. Tupin, "SAR-SIFT: A SIFT-like algorithm for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 453–466, Jan. 2015.

[28] X. Shen and W. Bao, "The remote sensing image matching algorithm based on the normalized cross-correlation and SIFT," *J. Indian Soc. Remote Sens.*, vol. 42, no. 2, pp. 417–422, Jun. 2014.

[29] H. Goncalves, L. Corte-Real, and J. A. Goncalves, "Automatic image registration through image segmentation and SIFT," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 7, pp. 2589–2600, Jul. 2011.

[30] H. Gonçalves, J. A. Gonçalves, and L. Corte-Real, "HAIRIS: A method for automatic image registration through histogram-based image segmentation," *IEEE Trans. Image Process.*, vol. 20, no. 3, pp. 776–789, Mar. 2011.

[31] Y. Ye, M. Wang, S. Hao, and Q. Zhu, "A novel keypoint detector combining corners and blobs for remote sensing image registration," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 3, pp. 451–455, Mar. 2021.

[32] X. Shen *et al.*, "RF-Net: An end-to-end image matching network based on receptive field," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8132–8140.

[33] J. Bian, W.-Y. Lin, Y. Matsushita, S.-K. Yeung, T.-D. Nguyen, and M.-M. Cheng, "GMS: Grid-based motion statistics for fast, ultra-robust feature correspondence," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 4181–4190.

[34] M. A. Fischler and R. C. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Commun. ACM*, vol. 24, no. 6, pp. 381–395, 1981.

[35] S. Wang, D. Quan, X. Liang, M. Ning, Y. Guo, and L. Jiao, "A deep learning framework for remote sensing image registration," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 148–164, Nov. 2018.

[36] Y. Tian, X. Yu, B. Fan, F. Wu, H. Heijnen, and V. Balntas, "SOSNet: Second order similarity regularization for local descriptor learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 11016–11025.

[37] L. Zhang and S. Rusinkiewicz, "Learning local descriptors with a CDF-based dynamic soft margin," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 2969–2978.

[38] S. Wang *et al.*, "Better and faster: Exponential loss for image patch matching," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4812–4821.

[39] D. Quan *et al.*, "AFD-Net: Aggregated feature difference learning for cross-spectral image patch matching," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3017–3026.

[40] V. S. Deshpande and J. S. Bhatt, "Bayesian deep learning for deformable medical image registration," in *Proc. Int. Conf. Pattern Recognit. Mach. Intell.* Cham, Switzerland: Springer, 2019.

[41] H. Li and Y. Fan, "Non-rigid image registration using self-supervised fully convolutional networks without training data," in *Proc. IEEE 15th Int. Symp. Biomed. Imag. (ISBI)*, Apr. 2018, pp. 1075–1078.

[42] B. D. de Vos *et al.*, "End-to-end unsupervised deformable image registration with a convolutional neural network," in *Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support*. Cham, Switzerland: Springer, 2017, pp. 204–212.

[43] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[44] S. Woo, J. Park, J. Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 3–19.

[45] Q. You, H. Jin, Z. Wang, C. Fang, and J. Luo, "Image captioning with semantic attention," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4651–4659.

[46] J. Lu, C. Xiong, D. Parikh, and R. Socher, "Knowing when to look: Adaptive attention via a visual sentinel for image captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 375–383.

[47] M. Pedersoli, T. Lucas, C. Schmid, and J. Verbeek, "Areas of attention for image captioning," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1242–1250.

[48] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[49] L.-C. Chen, Y. Yang, J. Wang, W. Xu, and A. L. Yuille, "Attention to scale: Scale-aware semantic image segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3640–3649.

[50] F. Wang *et al.*, "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 3156–3164.

[51] X. Chu, W. Yang, W. Ouyang, C. Ma, A. L. Yuille, and X. Wang, "Multi-context attention for human pose estimation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1831–1840.

[52] Y. L. Zhang, K. P. Li, K. Li, L. C. Wang, B. Zhong, and Y. Fu, "Image super-resolution using very deep residual channel attention networks," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 286–301.

[53] Y. Du, C. F. Yuan, B. Li, L. L. Zhao, Y. X. Li, and W. M. Hu, "Interaction-aware spatio-temporal pyramid attention networks for action classification," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 373–389.

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[54] J. Xu, R. Zhao, F. Zhu, H. Wang, and W. Ouyang, "Attention-aware compositional network for person re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2119–2128.
[55] W. Li, X. Zhu, and S. Gong, "Harmonious attention network for person re-identification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2285–2294.
[56] H. Zhang, I. Goodfellow, D. Metaxas, and A. Odena, "Self-attention generative adversarial networks," 2018, *arXiv:1805.08318*.
[57] M. Brown, G. Hua, and S. Winder, "Discriminative learning of local image descriptors," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 1, pp. 43–57, Jan. 2010.
[58] V. Balntas, K. Lenc, A. Vedaldi, T. Tuytelaars, J. Matas, and K. Mikolajczyk, "HPatches: A benchmark and evaluation of handcrafted and learned local descriptors," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 11, pp. 2825–2841, Nov. 2020.
[59] H. Gonçalves, J. A. Gonçalves, and L. Corte-Real, "Measures for an objective evaluation of the geometric correction process quality," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 2, pp. 292–296, Apr. 2009.
[60] C. A. Aguilera, F. J. Aguilera, A. D. Sappa, C. Aguilera, and R. Toledo, "Learning cross-spectral similarity measures with deep convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2016, pp. 1–9.
[61] C. A. Aguilera, A. D. Sappa, C. Aguilera, and R. Toledo, "Cross-spectral local descriptors via quadruplet network," *Sensors*, vol. 17, no. 4, p. 873, 2017.
[62] M. Brown and S. Susstrunk, "Multi-spectral SIFT for scene category recognition," in *Proc. CVPR*, Jun. 2011, pp. 177–184.
[63] *RGB-NIR Scene Dataset*. Accessed: Jun. 2011. [Online]. Available: https://www.epfl.ch/labs/ivrl/research/downloads/rgb-nir-scene-dataset/
[64] Y. Ye, L. Bruzzone, J. Shan, and L. Shen, "Fast and robust structure-based multimodal geospatial image matching," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 5141–5144.
[65] Y. Ye, L. Bruzzone, J. Shan, F. Bovolo, and Q. Zhu, "Fast and robust matching for multimodal remote sensing image registration," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9059–9070, Nov. 2019.

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